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Digital Circuits & Systems — Quick Revision Formula Sheet

Boolean Minimization, Setup/Hold Timing Constraints, Flip-Flop Conversions, ADCs & DACs

GATE Section: Section 5 • **Marks:** 8 - 10 Marks • **Core Devices:** Logic Gates, Counters, FSM, Converters

FLIP-FLOP EXCITATION & TIMING RESTRICTIONS
- **Clock Period Constraint:** T_clk >= t_cq + t_comb + t_setup - t_skew.
- **Hold Time Constraint:** t_cq + t_comb >= t_hold + t_skew (Hold violation is fatal; cannot be fixed by slowing clock!).
- **Characteristic Equations:** D-FF: Q+ = D | T-FF: Q+ = T xor Q | JK-FF: Q+ = J*Q' + K'*Q.

SECTION: BOOLEAN ALGEBRA & COMBINATIONAL ARITHMETIC

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
De Morgan's Laws	**(A + B)' = A' * B'** **(A * B)' = A' + B'**	Dual: AND <-> OR, 0 <-> 1	Universal gate count: 2-input XOR requires 4 NAND or 5 NOR gates.
Consensus Theorem	**A*B + A'*C + B*C = A*B + A'*C** **(A + B)*(A' + C)*(B + C) = (A + B)*(A' + C)**	Redundant term: B*C eliminated	Used to eliminate static hazards and glitches in combinational outputs.
2's Complement Range	**Range: -2^(n-1) to +2^(n-1) - 1**	For n = 8 bits: -128 to +127	Overflow in 2's complement: V = C_in xor C_out of MSB stage. Occurs when two positives add to negative or vice versa.
Full Adder Equations	**Sum = A xor B xor C_in** **C_out = A*B + B*C_in + A*C_in = A*B + (A xor B)*C_in**	Can be built with 2 Half Adders + 1 OR gate	Carry lookahead adder (CLA): Generate G_i = A_i*B_i, Propagate P_i = A_i xor B_i. C_i+1 = G_i + P_i*C_i.
Multiplexer Expansion	**N-to-1 MUX requires: 2^m inputs, m select lines**	Implements any (m+1) variable logic function without extra gates	Shannon's expansion theorem: f(w1, w2, ..., wn) = w1 * f(1, w2, ...) + w1' * f(0, w2, ...).

SECTION: SEQUENTIAL TIMING, COUNTERS & STATE MACHINES

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Setup & Hold Constraints	**Setup: T_clk >= t_cq + t_comb_max + t_setup** **Hold: t_cq + t_comb_min >= t_hold**	f_max = 1 / T_clk_min	Setup violations are fixed by reducing clock frequency. Hold violations cannot be fixed by clock frequency!
Clock Skew Effect	**Positive Skew (clock arrives late at destination):** **T_clk >= t_cq + t_comb + t_setup - t_skew** **t_cq + t_comb >= t_hold + t_skew**	Positive skew helps setup time, but hurts hold time!	Negative skew hurts setup time, reduces maximum operating frequency.
Ring & Johnson Counters	**Ring Counter: Mod-N using N flip-flops** **Johnson Counter: Mod-2N using N flip-flops**	Ring decoded states: N Johnson decoded states: 2*N (needs 2-input AND gates)	Johnson counter (twisted ring) feeds inverted Q_bar output of last FF back to first D input.

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Ripple Counter Delay	**Total propagation delay: t_total = N * t_pd**	f_max <= 1 / (N * t_pd)	Asynchronous counter: Output of one FF clocks the next. Fast for small N, severe glitching for large N.
Mealy vs. Moore FSM	**Mealy: Output = f(Present State, Current Input)** **Moore: Output = f(Present State only)**	Mealy reacts immediately to inputs; Moore output updates only on clock edge	Sequence detector for n-bit pattern: Moore requires (n + 1) states; Mealy requires n states.

SECTION: DATA CONVERTERS (DAC & ADC)

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
R-2R Ladder DAC	**V_out = -V_ref * sum_{i=1}^n [b_i / 2^i]**	b1 = MSB, bn = LSB	Requires only two precision resistor values: R and 2R. Output impedance is constant R.
DAC Resolution & Step Size	**Step Size = V_ref / (2^n - 1)** **Resolution (%) = 1 / (2^n - 1) * 100%**	Full Scale Output = V_ref * (2^n - 1) / 2^n ~ V_ref	For 8-bit DAC with 5V reference: Step size = 5V / 255 = 19.6 mV.
Flash ADC (Fastest)	**Comparators required: 2^n - 1** **Conversion time: 1 clock cycle (O(1))**	Resistors required in ladder: 2^n	8-bit Flash ADC needs 255 comparators! High power consumption and cost.
SAR ADC (Successive Approx)	**Conversion time: n clock cycles (independent of analog input magnitude!)**	Binary search algorithm using internal DAC and comparator	Standard industry ADC for microcontroller embedded systems.
Dual-Slope ADC (Highest Accuracy)	**Conversion time: (2^n + N_count) * T_clk**	Integrator slope: V_in / (R*C) for fixed T1, then V_ref / (R*C) down to zero	R and C values cancel out in final conversion equation! 50 Hz power line noise eliminated.